Module 3 · Functional Materials — MCQ Rapid Revision

Built only from your two decks: **A** = Module-3a Functional Materials (66 slides: electrical & magnetic) · **B** = Module-3b Nanomaterials & Quantum Dots (51 slides). "A21" = deck A, slide 21. Formulas are collected in §4; traps in §5.2; places where the slides contradict each other in §5.3.

memorise exactly (high-yield) trap / exception slides disagree — what to answer

grey italic = worked from a slide formula, not printed on a slide

1 · Electrical materials A1–A43

1.1 Ohm's law, resistance, resistivity A3–A5

- **Ohm's law:** $V = I \cdot R$ — current ∝ potential difference, **provided temperature & physical conditions stay constant**.
- **Valid only if:** temperature constant · material is **ohmic** (Cu, Al) · conductor not changing shape/length.
- **Non-ohmic devices** (do NOT obey): diodes, transistors.
- Ohmic V–I graph = **straight line**.
- **Resistance (R):** property of a conductor that **opposes** current. R = V/I and $R = \rho L/A$.
- **Longer wire → more R** (longer path, more collisions with atoms); **thicker wire → less R** (more room for electrons).
- **Resistivity (ρ):** intrinsic property of a material — its "**fingerprint**" for conduction; = resistance normalised to unit length & unit area. ρ = RA/L = VA/(IL), unit **Ω·m**.
- Depends only on material and temperature; **not** on shape or size (Cu has low ρ however long/thick).
- Low ρ → good conductors (Cu, Ag); medium → semiconductors (Si, Ge); high → insulators.

Type (A5 table)	Resistivity range (Ω·m)
Conductors	$10^{-8} – 10^{-6}$
Semiconductors	$10^{-5} – 10^{6}$
Insulators	$10^{8} – 10^{16}$

Trap: "Resistivity of a wire doubles when its length doubles" — FALSE. **Resistance** doubles; **resistivity** is unchanged (material + temperature only).

Resistivity values A7

Material	ρ (Ω·m)	Material	ρ (Ω·m)
Silver	1.6×10^{-8}	Silicon	$\sim 10^{3}$
Copper	1.7×10^{-8}	Glass	$10^{10} – 10^{14}$
Aluminium	2.7×10^{-8}	Rubber	$\sim 10^{13}$
Iron	1.0×10^{-7}		

Low-ρ metals, all with positive α A20

Metal	ρ (μΩ·cm)	α (/°C)
Silver	1.58	0.0038
Copper	1.68	0.0037
Gold	2.21	0.0034
Aluminium	2.65	**0.00429** (highest)

Order of ρ: Ag < Cu < Au < Al. 1 μΩ·cm = 10^{-8} Ω·m.

1.2 Six factors affecting resistivity A6–A10, A19

Factor	Effect on ρ
1 Nature of material	Decided by the **number of free electrons**. Many (Ag, Cu, Al) → very low ρ ($\sim 10^{-8}$ Ω·m); very few/none (rubber, wood, glass) → very high (10^{10}–10^{14} Ω·m); semiconductors intermediate, **controllable by temperature and impurities**.
2 Temperature	**Metals: ρ ↑** as T ↑ (atoms vibrate more → more collisions/scattering). **Semiconductors & insulators: ρ ↓** as T ↑ (more electrons gain energy to jump to the conduction band).
3 Impurities	**Metals:** impurities increase electron scattering → **ρ ↑** (pure Cu has lower ρ than **brass** = Cu + Zn). **Semiconductors:** impurities added deliberately (**doping**) → **ρ ↓ drastically** (pure Si high ρ; + phosphorus → n-type, + boron → p-type) — basis of diodes, transistors, ICs.
4 Mechanical strain / stress	Localised strains in the crystal disturb free-electron movement → **ρ ↑**.
5 Phase changes	Change of physical state (solid/liquid/gas) or crystal phase can **drastically** change ρ (electron motion depends on structure).
6 Crystal defects	Perfect lattice → low ρ; defects scatter electrons → **ρ ↑**. Annealing (heating + slow cooling) reduces defects → **ρ ↓**.

Exception: in a metal, any impurity increases ρ — even an impurity whose own resistivity is lower, because it creates lattice imperfections (A19). In a semiconductor, the right impurity **decreases** ρ.

1.3 Temperature coefficient α A8, A20

- $\rho_T = \rho_0(1 + \alpha\Delta T)$, $\Delta T = T - T_0$ (full symbols in §4).
- α = fractional change in resistance per degree change in temperature (/°C).
- α > 0 for metals; α < 0 for semiconductors AND insulators.
- Metals with **high α** → **not** suitable for precise resistors.
- Alloys with **low α** (**Constantan, Nichrome**) → standard resistors, heating coils. Nichrome's very small α → **heating elements**.
- Semiconductors (**negative α**) → thermistors (temperature sensors).

1.4 Energy band theory A12–A13

Isolated atom → **discrete** levels. In a solid, atoms are close, **outer orbitals overlap** → each level splits into many closely spaced levels → **energy bands**. In solids we talk about bands, not single levels.

	Valence band (VB)	Conduction band (CB)	Forbidden gap (E_g)
What	Levels of outermost (valence) electrons; lower energy; lies **below** CB	Levels above VB; electrons here are **free**, not bound to the nucleus	Energy difference between CB and VB; **no allowed electron states**
At 0 K	Usually **completely filled**; electrons loosely bound, not free unless excited	**Empty** in semiconductors & insulators; partially filled / overlapping VB in conductors	—
Role	Supplies electrons	Enables conduction	**Decides material type**

1.5 Conductors vs semiconductors vs insulators A13, A15–A17

Property	Conductor	Semiconductor	Insulator
Current	Allows very easily	Allows partially	Does not allow
Band picture	VB & CB **overlap**, OR CB partially filled	Small gap	Large gap
E_g	≈ 0 eV	0.7–1.5 eV (~1 eV); A17 table: "< 5 eV"	> 5 eV
Free electrons	Large number, always available	Few	None (almost)
Energy to go VB → CB	Very little	High	Very high
Conductivity	Very high	Moderate; **increases with T**; raised by doping	Negligible / "nil"
Valence electrons	< 4	= 4	> 4
Temp. coeff. of resistance	**Positive**	**Negative**	**Negative**
At 0 K	Conducts	VB full, CB empty → **behaves as insulator**	Insulator; a few e^- cross only at very high T or strong field
Examples	Cu, Ag, Al, W	Si, Ge, Se, C	Diamond, rubber, glass, wood, paper, plastics

Trap: carbon is listed as a semiconductor example (A17), yet **diamond** (carbon) is an insulator (E_g ~5.5 eV, A28) and **graphite** is diamagnetic (A52). Read the form of carbon in the question.

1.6 Conductors (low-resistivity materials) A19–A27

Used for: windings of electrical machines/apparatus; transmission & distribution of electrical energy.

Required properties (8): highest possible conductivity · least temperature coefficient of resistance (ideally zero) · high melting point · high mechanical strength · high ductility (drawn into wire) · high corrosion resistance (no oxidation) · solderability · long life, durable, flexible.

Copper A21

- Configuration [Ar] $3d^{10}$ $4s^1$; the $4s^1$ electron is the valence electron.
- Close packing → 3d and 4s overlap → partially filled bands; slide: "VB (4s) and CB (3d) overlap" → **$E_g \approx 0$**; free electrons always available.
- High electron mobility (fewer scattering events). Conductivity second only to silver (one free electron per atom + overlapping bands).
- $\rho \approx 1.7 \times 10^{-8}$ Ω·m at 20 °C; good thermal conductor (free electrons carry heat); ductile & malleable.
- **Uses:** wiring & power cables, PCBs, contacts & connectors, heat exchangers.

Aluminium A25–A27

- [Ne] $3s^2$ $3p^1$ → 3 valence electrons.
- **3rd most abundant element** (after O, Si) and **most abundant metal** in Earth's crust.
- $\rho \approx 2.7 \times 10^{-8}$ Ω·m (20 °C); density 2.7 g/cm^3 (Cu ≈ 8.9 g/cm^3).
- VB & CB overlap → many free electrons. α positive.
- Conductivity ≈ 61 % of copper (same cross-section). A larger-diameter Al conductor gives the same conductivity as Cu at **half the weight**.
- Thin **Al_2O_3** oxide layer → corrosion resistant outdoors.
- Weaker than steel → ACSR = Aluminium Conductor Steel Reinforced (Al for conductivity + steel core for strength).
- **Uses:** overhead transmission lines (**most common**), bus bars in substations, power cables where weight matters; house wiring less common now (**Cu preferred for safety**).

Property A23–24	Hard-drawn Cu	Annealed Cu
Made by	Cold working / wire drawing, **no heating**	Heating hard-drawn Cu, then **slow cooling**
Microstructure	High dislocation density, internal stresses	Fewer dislocations, relaxed grains
Tensile strength	**Higher**	Lower
Ductility / hardness	Lower, harder	**Higher**, softer
Conductivity	Slightly lower (defect scattering)	**Higher**
Uses	Overhead lines (strength)	House wiring, cables (flexibility)

Property A27	Copper	Aluminium
ρ at 20 °C	~1.7 × 10^{-8} Ω·m	~2.7 × 10^{-8} Ω·m
Conductivity	Higher	Lower (~61 % of Cu)
Density	8.9 g/cm^3	2.7 g/cm^3
Weight (same length & area)	Heavy	~⅓ of copper
Cost	Expensive	Cheaper
Uses	House wiring, PCB, cables	Overhead lines, bus bars

Trap: Al weighs **~⅓ of Cu for the same size** (A27) but **~½ of Cu for the same conductivity** (thicker wire, A25). Both are correct — check which is asked. Al's ρ (2.7 × 10^{-8}) and density (2.7 g/cm^3) share the digits 2.7.

1.7 Insulators A28

- VB completely filled, CB empty, **E_g > 5 eV** → electrons cannot reach the CB normally.
- **Band gaps:** diamond ~5.5 eV · glass ~9 eV · PVC, rubber ≫ 6 eV.
- **Properties:** ρ 10^8–10^{16} Ω·m; negligible conductivity; **negative** temperature coefficient; electrical isolation; strong **dielectrics** (withstand high voltage without breakdown).
- **Materials:** non-metals – rubber, glass, plastic, wood, **mica**; ceramics – **porcelain, alumina, SiO_2**; others – PVC, **Bakelite**, paper (oil-impregnated, for cables).

Application	Insulator used
Wiring insulation	Plastic coatings, PVC, rubber
Transmission-line pylons	Porcelain or glass insulators
Transformers & capacitors	Mica, ceramics, impregnated paper
PCBs	Epoxy resins, fiberglass
High-voltage equipment	Ceramic insulators, vacuum, SF_6 gas

1.8 Semiconductors A30–A33

- Conductivity between conductors and insulators; **insulator at 0 K**, conducts at room temperature because E_g is small (0.7–1.5 eV) — some electrons gain thermal energy and cross to the CB.
- **Charge carriers:** electrons in CB (**negative**) + **holes in VB** (absence of electron, **positive**) → current carried by **both**.
- **Elemental:** Si, Ge. **Compound:** GaAs, InP.
- **Intrinsic (pure Si, Ge):** conduction only from **thermally generated electron–hole pairs**.
- **Extrinsic (doped):** doping ≈ 1 impurity atom per 10^8 atoms, to increase conductivity.
- **Compound** (≥ 2 elements; high-frequency & optoelectronic): GaAs → LEDs, laser diodes, microwave ICs; InP → high-speed electronics, optical communication.

	n-type	p-type
Impurity	**Pentavalent** (Group V)	**Trivalent** (Group III)
Examples	P, As, Sb (A32 adds **Bi**)	B, Al, Ga, In
Impurity e^-	5 — one extra	3 — one short
Called	**Donor**	**Acceptor**
Majority / minority	Electrons / holes	Holes / electrons
Conduction mainly by	Electrons	Holes
Impurity becomes	Positively ionised (donates e^-)	Negatively ionised (accepts e^-)
Slide example	Si + phosphorus	Si + boron

Trap: the ion charge is the **opposite** of the type letter — in **n**-type the donor ion is **+**; in **p**-type the acceptor ion is **–**.

Area A33	Devices → uses
Electronics	Diodes → rectifiers, voltage regulators · Transistors → amplifiers, switches · ICs → microprocessors, memory chips
Power devices	**SCR** (Silicon Controlled Rectifier), **IGBT** (Insulated Gate Bipolar Transistor), **Power MOSFET** (Metal-Oxide-Semiconductor Field-Effect Transistor) → converters, inverters, motor drives
Optoelectronics	LEDs (GaAs) → displays, indicators · Laser diodes (GaAs) → optical communication, CD/DVD · Photodiodes, solar cells → light detection, renewable energy
Comm. & computing	GaAs & InP chips → high-speed computers, 5G/6G (InP) · Sensors → temperature, pressure, infrared, biosensors
Everyday	Phones, computers, TVs, digital cameras, smart appliances

1.9 Superconductors A34–A43

Superconductor: material that, cooled below a **critical temperature T_c**, loses all electrical resistance → current flows with no energy loss. A **quantum-mechanical** phenomenon with two signatures: **(1) zero resistance below T_c, (2) perfect diamagnetism (Meissner effect)**.

Year	Event A34, A36, A40
1911	**Heike Kamerlingh Onnes** (Dutch) discovers superconductivity: resistance of solid **mercury vanishes at 4.2 K**, using liquid helium
1913	Onnes wins Nobel Prize in Physics — **first to liquefy helium** (cooled to ~1.5 K, then the coldest on Earth)
1933	**Meissner & Ochsenfeld**: superconductors expel magnetic field; repulsion can levitate them
1986	**Bednorz & Müller**: high-temperature superconductors (HTS), T_c above 77 K (per A34)
1987	Bednorz & Müller Nobel Prize in Physics; **YBCO** discovered ($T_c \approx 92$ K)
Boiling point of liquid N_2 = −195.795 °C = 77.355 K	

- **Zero resistivity (A35):** normal metal → resistance falls **gradually** as T is lowered (levels off at a finite value); superconductor → **drops suddenly to zero at T_c**.
- $\rho(T) = \rho_0$ for $T > T_c$; $\rho(T) = 0$ for $T < T_c$.
- **Persistent current:** flows indefinitely without power loss.
- **Meissner effect (A36):** superconductor **expels magnetic flux lines** when cooled below T_c. Little/no field inside = **Meissner state**. It distinguishes superconductors from perfect conductors. $\chi = -1$ (perfect diamagnet).
- **Critical field H_c:** max external field at a given T before returning to the normal (resistive) state; $H > H_c(T)$ destroys superconductivity. Type I: $H_c(T) = H_c(0)[1 - (T/T_c)^2]$ (§4).
- **Critical current density J_c:** maximum current density before breakdown.

	Type I (soft)	Type II (hard)
Meissner	**Complete**	**Partial** flux penetration (vortices)
Critical fields	One, **low**	**Two:** H_{c1} = flux **starts** penetrating; H_{c2} = superconductivity **destroyed**
Transition	Sharp normal ↔ superconducting	—
Current capacity	—	**High**
Examples	Pb, Hg, Al	NbTi, Nb_3Sn, $YBa_2Cu_3O_7$ (YBCO/Y123)

Material A39	T_c (K)	Type
Mercury (Hg)	4.2	I
Lead (Pb)	7.2	I
Niobium (Nb)	9.2	**II**
NbTi alloy	10	II
Nb_3Sn	18	II
$YBa_2Cu_3O_7$ (YBCO)	92	II (HTS)
$Bi_2Sr_2Ca_2Cu_3O_{10}$ (BSCCO)	~110	II (HTS)

YBCO — $YBa_2Cu_3O_7$, yttrium barium copper oxide A40

- **Ceramic oxide** superconductor, discovered **1987**; **cuprate** family; **perovskite-like layered** structure.
- Contains: yttrium (Y) layer · barium (Ba) layers · CuO_2 planes — where superconductivity occurs · oxygen ions controlling charge balance and **hole concentration**.
- **Oxygen deficiency (δ) changes T_c**. Type II; shows Meissner effect below T_c.

- $T_c \approx 92$ K · $H_{c2} > 100$ T · $J_c \sim 10^6$ A/cm^2
- **First material with $T_c > 77$ K** → cooled by **liquid nitrogen instead of liquid helium** → cheaper, practical.
- Triggered the HTS revolution; reduced cost and complexity of superconducting applications.

Area A42–A43	Applications of superconductors
Power & energy	Power cables (no losses; **Essen, Germany**) · generators & motors (smaller, efficient; naval propulsion, industry) · **Fault current limiters** (limit grid surges instantly) · **SMES** (stores energy as a **magnetic field**; instant backup)
Medical	**MRI** (strong, stable fields; replaced resistive magnets) · **NMR** spectroscopy (molecular structure)
Transport	**Maglev** (levitate + propel; **Japan SCMaglev > 600 km/h** in tests) · compact magnets for spacecraft propulsion
Electronics & computing	Josephson junction = two superconductors separated by a thin non-superconducting barrier (thin insulator or metal) → superfast switches; **RSFQ** (Rapid Single Flux Quantum) logic · SQUID = superconducting loop with two Josephson junctions; detects **femtotesla** fields → brain mapping (MEG), mineral exploration, submarine detection · superconducting **qubits** (Google, IBM)
Research	**LHC (CERN)** magnets guide protons; superconducting RF cavities accelerate particles · **fusion** (tokamaks, ITER) magnets confine plasma
Misc.	Levitation demos on YBCO pellets · sensitive detectors (infrared, astronomical telescopes) · military radar, naval propulsion

2 · Magnetic materials A44–A66

2.1 Origin of magnetism, magnetic moment A45–A48

- Magnetism arises from the **motion of electrons** within atoms. **All** materials are magnetic to some extent; behaviour is set by **electron configuration**, especially **unpaired electrons**. Materials are classified by their response to an external field (very weak → very strong).
- **Two sources:** (1) **orbital** motion around the nucleus = tiny current loops → **orbital magnetic moment**; (2) electron **spin** (intrinsic angular momentum) → **spin magnetic moment**.
- Net atomic moment = vector sum of orbital + spin contributions. Magnetic dipole moment is a **vector** (magnitude + direction) = strength & direction of magnetism. Many aligned dipoles → a magnetic field.

- **Bohr magneton** μ_B = fundamental unit of atomic moment: $\mu_B = e\hbar/2m_e = eh/4\pi m_e = 9.27 \times 10^{-24}$ A·m^2 (J/T).
- Spin g-factor (Landé) $g_s = 2$. Orbital & spin formulas → §4.
- **Spin-only moment** (orbital part quenched in crystals for most transition-metal ions): $\mu_{eff} = \sqrt{(n(n+2))}\ \mu_B$, n = unpaired electrons.

n unpaired	0	1	2	3	4	5
μ_{eff} (μ_B)	0	1.73	2.83	3.87	4.90	5.92

Values calculated from the slide formula.

2.2 Susceptibility χ and permeability μ A49–A50

- **Susceptibility:** how much a material becomes magnetised in an applied field; tells whether it is **attracted (χ > 0)** or **repelled (χ < 0)**. $\chi = M/H$.
- **M** = intensity of magnetization = **magnetic moment per unit volume**; M ∝ H. **H** = applied magnetic field strength (A/m).
- **Permeability:** how easily a material supports formation of a magnetic field within itself (ability to conduct magnetic lines of force). $\mu = B/H$; B in tesla (T), H in A/m, μ in H/m.
- $\mu_0 = 4\pi \times 10^{-7}$ H/m (free space). $\mu_r = \mu/\mu_0$.
- $\mu_r = 1 + \chi$; $\mu = \mu_0(1 + \chi)$ — valid for **linear** (B = μH, directly proportional) and **isotropic** (same properties in all directions) materials.
- **Anisotropic** = directional preference; crystalline materials can be anisotropic; perfect isotropy is an idealisation.
- *Check: $\chi < 0 \Rightarrow \mu_r < 1$; superconductor $\chi = -1 \Rightarrow \mu_r = 0$.*

2.3 The five classes A47, A51–A64

Class	Spins / cause	χ	μ_r	In a field	Temperature	Examples
Diamagnetic	**All electrons paired** → no net moment; field induces a weak moment **opposing** it	< 0, small	Slightly < 1	Weakly **repelled**; lines of force repelled	Independent of T	Bi, Cu, Ag (A51); Bi, Cu, Au (A54)
Paramagnetic	**Unpaired** electrons; moments **random** with no field; align weakly with field; alignment temporary, disrupted by thermal energy	> 0, small	Slightly > 1	Weakly **attracted**	Curie law $\chi = C/T$: T ↑ → χ ↓	Al, O_2, Mg (A51); Al, Pt, Na, O_2 (A55)
Ferromagnetic	Moments **parallel** within **domains** (exchange interactions)	≫ 1	10^2–10^5 or more	Strongly attracted; **retains magnetization** after field removed; **hysteresis**	Spontaneous magnetization below T_C; paramagnetic above (Curie–Weiss)	Fe, Co, Ni (A51); + Gd (A58)
Anti-ferromagnetic	Neighbouring moments **antiparallel, equal** → cancel	> 0, small	≈ 1	Net magnetization ≈ 0	χ **rises** to a **peak at T_N**, then falls; paramagnetic above T_N	MnO, FeO
Ferrimagnetic	Antiparallel, **unequal** → do not cancel	> 0, moderate–large	10^2–10^4	Net magnetization ≠ 0	Ordered below T_c (A64)	Fe_3O_4 (magnetite), ferrites

Example lists A52 — favourite source for "which is …?" MCQs	
Diamagnetics	Copper, mercury, bismuth, zinc, magnesium, gold, silicon, phosphorus, **water**, **graphite**, proteins
Paramagnetics	Platinum, aluminium, **air**, sodium, tin, oxygen, hydrogen
Ferromagnetics	Iron, cobalt, nickel, **gadolinium, dysprosium, erbium, holmium**
Antiferromagnetics	MnO, FeO, CoO, FeMn, NiO, MnS, MnF_2
Ferrimagnetics	Fe_3O_4, $PbFe_{12}O_{19}$, $BaFe_{12}O_{19}$, $BaO{\cdot}6Fe_2O_3$, $SrO{\cdot}6Fe_2O_3$

Iron-family trap: **Fe** = ferromagnetic · **FeO** = antiferromagnetic · **Fe_3O_4** = ferrimagnetic. **Cu, Ag, Au** are diamagnetic even though they are excellent conductors; **Al** and **Pt** are paramagnetic.

Ferromagnetism & Curie temperature A56–A60

- Strong, **permanent** magnetism from collective alignment of atomic moments in microscopic **magnetic domains** (all moments in a domain aligned).
- Unmagnetised: domains randomly oriented. External field lines domains up → strong magnetization. It is the **basic mechanism of permanent magnets** — magnetic even without a field.
- Whether a compound can be ferromagnetic depends on its **number of unpaired electrons** and its **atomic size**.
- **Curie temperature T_C**: above it a ferromagnet loses ferromagnetism and becomes paramagnetic. Below T_C: exchange interactions → spontaneous magnetization, very high χ. **At T_C:** thermal energy breaks alignment; χ diverges (→ ∞). Above T_C: Curie–Weiss law $\chi = C/(T - T_C)$.
- **Hysteresis:** magnetization depends on previous magnetic history.

Antiferromagnetism & Néel temperature A61

- Zero or very small net macroscopic magnetization.
- χ **increases** with temperature, reaches a **maximum at the Néel temperature T_N**, then decreases; above T_N the material is paramagnetic.

χ–T graph shapes

- **Para:** χ vs T falls as a curve ($\chi = C/T$); 1/χ vs T = straight line through origin, slope 1/C.
- **Ferro:** χ very high, falls steeply to T_C, then a paramagnetic tail.
- **Antiferro:** rising curve → **peak (cusp) at T_N** → paramagnetic fall.
- **Dia:** flat — independent of T.

Spin diagrams A47, A63–64

- Field ON: dia → small magnetization **opposite** field; para → small, **parallel**; ferro → large, parallel; antiferro → small; ferri → net parallel.
- Field OFF: dia & para → none (para spins random); **ferro & ferri keep magnetization**; antiferro → zero.

T_C (Curie) and T_N (Néel) values, K A62			
Element	**T_C**	**T_N**	**Note**
Co	1404	—	highest T_C
Fe	1043	—	
Ni	631	—	
Gd	293	—	≈ room temp.
Tb / Dy	222 / 85	229 / 179	has both; $T_N > T_C$ in every case
Ho / Er	20 / 20	131 / 84	
Tm	25	56	
Cr / Mn	—	308 / 100	T_N only (antiferro)
Ce / Pr / Nd / Sm / Eu	—	12.5 / 25 / 19 / 14.8 / 90	

Applications A65

- **Data storage:** magnetic tapes, hard disks, credit-card strips use the **memory effect of ferromagnetic** materials.
- **Electronics:** soft magnetic materials (high permeability, low energy loss) → cores of **transformers, motors, inductors**.
- **Permanent magnets:** hard magnetic NdFeB, SmCo → motors, speakers, generators.
- **Medical:** MRI (superconducting magnets); **paramagnetic contrast agents**; **superparamagnetic nanoparticles** for targeted drug delivery.
- **Transport:** Maglev uses strong repulsion of superconductors ("a type of diamagnetic material").
- **Sensors:** Hall-effect, magnetoresistive. **Shielding:** mu-metal (soft, high permeability).

Storage devices A66

- **HDD:** magnetises small regions of a disk; **domain orientation = binary 0/1**. Tapes & floppy disks — same principle.
- **Magneto-optical:** laser heating + magnetic field; heated locally above the Curie temperature → magnetization reversible → rewriting.
- **MRAM** (Magnetoresistive RAM): orientation of magnetic layers stores data; non-volatile.
- Uses the concepts of susceptibility and Curie temperature.

Symbol trap: T_c = **critical** temperature for superconductors (A34–40) but T_C = **Curie** temperature for ferromagnets (A58). Read the context. T_N = Néel (antiferromagnets).

3 · Nanomaterials & quantum dots B1–B51

3.1 Definition, scale, where used B2–B7

- **Nanoscale materials:** at least ONE dimension < ~100 nm. Unique optical, magnetic, electrical properties emerge at this scale.
- 1 nm = 10^{-9} m = one millionth of a millimetre (a billionth of a metre).
- Atom ≈ 0.1 nm → 10 atoms side by side = 1 nm.
- **Human hair:** 50,000–100,000 nm diameter; a nanometre is ~100,000× smaller than a hair's diameter.
- Scale (B4): biomolecules (DNA, antibody, glucose, protein, virus) 1–100 nm · bacteria ~10^3 nm · cancer cell ~10^4 · pencil tip ~10^6 · tennis ball ~10^8 nm. Nanostructures shown: fullerene, dendrimer, quantum dot, gold, metal oxide (Fe_3O_4), liposome, polymer.
- Nanomaterials are the cornerstones of nanoscience & nanotechnology — a broad, interdisciplinary, fast-growing field already with commercial impact.
- Some occur **naturally**; of special interest are **engineered nanomaterials (EN)**.
- **Products:** sunscreens, cosmetics, sporting goods, stain-resistant clothing, tyres, electronics; medicine (diagnosis, imaging, **drug delivery**); nanocoatings/nanocomposites in windows, sports equipment, bicycles, cars.
- Nano-TiO_2 → cosmetics, sun-block creams, **self-cleaning windows**.
- Nano-silica → filler in cosmetics and **dental fillings**.
- **Butyl-rubber / nano-clay** composite → longer-lasting tennis balls. UV-blocking coatings on glass bottles protect drinks from sunlight.
- **Fields (B7):** IT, mechanical engineering/robotics, biotechnology, transportation, national security & defence, food & agriculture, medicine/health, aerospace, energy & environment, advanced materials & textiles.

3.2 Classification B8, B10

Class	Dimensions	Examples
0D	**All three** in nanoscale	Nanoparticles, colloids, quantum dots
1D	**One** beyond nanoscale, two in it	Nanowires, nanorods, nanotubes, biopolymers
2D	**Two** outside, one in nanoscale	Plate-like nanolayers, surface coatings, thin films
3D	All three can be outside; **made of** nanomaterials	Nanoparticle dispersions, nanowire/nanotube bundles, multiple nanolayers

Rule: the "D" counts dimensions **outside** (free of) the nanoscale — so a quantum dot is **0D**, a thin film is **2D**.

Category	Examples
Carbon-based (organic)	Fullerenes, carbon nanotubes
Inorganic – metal oxides	ZnO, iron oxide, cerium oxide
Inorganic – metals	Gold, silver, iron
Inorganic – quantum dots	CdS, CdSe
Polymeric	(polymer nanomaterials)

3.3 Why nanomaterials are interesting and different B11, B17–B19

- Unusual **mechanical, electrical, optical, magnetic** properties. Examples: **nanophase ceramics are more ductile at elevated temperatures** than coarse-grained ceramics · **nanostructured MnO_2** → rechargeable batteries (cars, consumer goods) · very small particles have **discrete electronic states** → special properties plus **superparamagnetism**.
- Two main reasons for different properties: (1) increased relative surface area, (2) new quantum effects.
- **Surface area / volume:** subdividing keeps total volume the same but greatly increases collective surface area → **better catalytic efficiency**.

1 cm^3 cube cut into	Cube edge	Total area
1 cube	1 cm	6 cm^2
8 cubes	½ cm	12 cm^2
27 cubes	⅓ cm	18 cm^2

Pattern: edge 1/n cm → n^3 cubes, area 6n cm^2 (e.g. n = 10 → 60 cm^2).

- **Other changes (B19):** fewer imperfections/defects → changed electrical conductivity; more structural perfection → **higher mechanical strength** (hardness, toughness); increased **paramagnetism and "supermagnetism"** behaviour; smaller size → **higher permeability through biological membranes**.

Quantum size effect / quantum confinement B18, B42, B44

- Particles so small that electrons are **not free to move** as in bulk gold → they interact differently with light → **tunable fluorescent emission**.
- Electronic states become **discrete energy levels**; confinement increases the energy gap → metal → semiconductor → insulator transition. Size ↓ ⇒ band gap ↑.
- **Quantum confinement** = spatial confinement of electrons in a nanomaterial (instead of free existence in the bulk). Electron–hole pairs become confined when the particle diameter approaches the de Broglie wavelength of conduction-band electrons, **λ = h/mv**.

3.4 Bulk gold vs nano gold; colours B12–B15

Bulk gold	Nano gold
Lustrous (shiny when polished)	Never gold-coloured; a range of colours
Malleable; ductile (drawn into wires)	Colour set by particle **size & shape**
Metallic, yellow in mass	Not metal — semiconductor, band gap 3.4 eV
Good conductor of heat & electricity	Very good catalyst
Melting point ~1080 °C (high)	Melts lower, ~940 °C
Bulk only: high density; hard & tough, high tensile strength; resists stretching/drawing stresses; not easily breakable; **inert** (unaffected by air & most reagents)	

In glass B13	Size	Shape	Colour reflected
Gold	25 nm	sphere	Red
	50 nm	sphere	Green
	100 nm	sphere	Orange
Silver	40 nm	sphere	Blue
	100 nm	sphere	Yellow
	100 nm	**prism**	Red

- **Surface plasmon resonance (SPR)**: light's electromagnetic field interacts with the collective oscillation of electrons on the nanoparticle surface → some wavelengths absorbed, others scattered/reflected → colour. Colour depends on **size, shape, material**, and also **distance between particles**.
- B14 note: "nanomaterials scatter visible light rather than absorb".

3.5 Synthesis: top-down vs bottom-up B20–B24

	Top-down	Bottom-up
Principle	**Breaking down** larger pieces of material	**Assembling** single atoms & molecules into larger nanostructures
Path	Bulk → powder → nanoparticles	Atoms/molecules → nuclei & growth (clusters) → nanoparticles
Best for	**Long-range order**, macroscopic connections	**Short-range order** at nanoscale dimensions, assembly
Nano-silver example	Bulk piece **mechanically ground**	**Chemical and biological** methods
Methods	Mechanical milling · etching · laser ablation · sputtering · electro-explosion	Supercritical fluid synthesis · spinning · sol-gel · laser pyrolysis · chemical vapour deposition · molecular condensation · chemical reduction · green synthesis
Slide example	Mechanical milling	Sol-gel

Look-alikes: laser **ablation** = top-down, laser **pyrolysis** = bottom-up · **sputtering** = top-down, **spinning** = bottom-up.

Mechanical milling (top-down) B25–B31

- Also called mechanical alloying / mechanochemical synthesis. High-energy mills repeatedly impact and deform powder → **size reduction + new phases**.
- Makes **non-equilibrium structures**: nanocrystalline, amorphous, quasicrystalline.
- Feed powder ~50 µm + balls of **hardened steel or tungsten carbide (WC)-coated** (refractory or steel) in a **sealed** container, shaken/violently agitated; preferably **inert atmosphere**.
- Most effective ball : powder mass ratio = 5 : 10.
- **Mill types:** tumbler, attrition, shaker, vibratory, planetary. **Shaker mill** e.g. **SPEX 8000** → small batches (~10 cm^3 powder). Planetary: bowl rotation + supporting disc, centrifugal force.
- **On impact:** elastic deformation → plastic & shear deformation → fracture, amorphization, chemical reactions.
- **Uses:** cost-effective; blends of phases → **nanocomposites**; oxide- and carbide-strengthened Al alloys; Al/Ni/Mg/Cu-based nanoalloys; **ball-milled carbon nanomaterials** (novel class) → environmental remediation, energy storage, energy conversion.
- **Advantage:** high production rates.
- **Disadvantages:** severe plastic deformation — interface heats to 100–200 °C · hard to reach required particle size · **contamination** from milling tools (Fe) and atmosphere (trace O_2, N_2 in rare gases) → needs inert conditions, e.g. glove box (Fe < 1–2 %, trace elements < 300 ppm) · protective coatings against milling-tool contamination (MTC) raise cost · > 30 h milling → MTC > 10 %.

Sol-gel reactions (Si example) B34–B35

Reaction	Equation	Reverse
Hydrolysis	≡Si–OR + H_2O → ≡Si–OH + ROH $Si(OR)_4 + 4H_2O \rightarrow Si(OH)_4 + 4ROH$	Re-esterification
Water condensation	≡Si–OH + HO–Si≡ → ≡Si–O–Si≡ + **H_2O**	Hydrolysis
Alcohol condensation	≡Si–OR + HO–Si≡ → ≡Si–O–Si≡ + **ROH**	Alcoholysis

Si–O–Si links = **siloxane bridges** → network of siloxanes. General: –MOR + H_2O → –MOH + ROH.

Sol-gel (bottom-up) B32–B40

- For high-quality **metal-oxide** nanomaterials. Liquid precursor → **sol** (solid particles in solution) → **gel** (network). Precursors: metal alkoxides (drawn as $Si(OC_2H_5)_4$).
- **Steps:** hydrolysis → condensation → aging → drying → calcination.
  ① **Hydrolysis** of alkoxide in water or with alcohol → sol. ② **Condensation/polycondensation**: viscosity ↑, porous structure; hydroxo (M–OH–M) or oxo (M–O–M) bridges → metal-hydroxo/oxo polymers. ③ **Aging**: polycondensation continues; porosity decreases, distance between colloidal particles increases. ④ **Drying**: water & organic solvents removed. ⑤ **Calcination** → nanoparticles.
- **Flow (B36):** precursor + solvent → transparent solution → (hydrolysis & polymerization) sol → gel → speed drying → **xerogel** → calcination → nanopowder.
- **Products from a sol (B37):** coating → xerogel film → heat → dense film · gelling → wet gel → evaporation → xerogel → heat → dense ceramic · wet gel → solvent extraction → **aerogel** · precipitating → uniform particles · spinning → furnace → ceramic fibres.
- Aerogel = dried by supercritical extraction; xerogel = dried by ordinary evaporation at room temperature. (Supercritical CO_2 details: B38 says **not in syllabus**.)
- **Advantages:** nonmetallic inorganic materials (glasses, glass ceramics, ceramics) at **very low temperatures** vs melting/firing · **monosized nanoparticles**.
- **Disadvantages:** controlling growth and stopping **agglomeration** · hard to ensure complete reaction · completely removing growth aids · slow production rate.

ZnO nanoparticles by sol-gel B39

- NaOH **8 g** + 10 mL distilled water, stir 5 min.
- Zinc acetate dihydrate **2 g** + 15 mL distilled water, stir 5 min.
- Add NaOH solution into zinc acetate solution, stir 5 min.
- Add ethanol (100 mL) drop-wise from a burette → **white precipitate**.
- Centrifuge **15 min at 5000 rpm** → ZnO NPs.

> **Contrast:** milling = **high** production rate; sol-gel = **slow** production but **monosized** particles at **low** temperature.

3.6 Quantum dots (QDs) B41–B51

- **Semiconductor particles**, 2–10 nm diameter ("< 10 nm", B51); named so because quantum effects dominate their light emission. **0D** nanomaterial.
- Examples: CdSe, CdS, InP. (Carbon dots also appear in synthesis slides.)
- **Emission mechanism:** external stimulus → some electrons absorb enough energy to escape their atomic orbit → a conductance region (they conduct) → electrons drop back → energy released as **light**; colour = amount of energy released.
- Contain 100–1000 electrons; 2–10 nm ≈ 10–50 atoms in diameter.
- **Tunable band gap** is the most attractive feature → flat-panel displays, photovoltaics, optoelectronic devices, lasers, sensors, photon-bandgap devices.
- **Properties:** high extinction coefficient · high electron mobility · high bandwidth & position adjustment · high solution-process capability (repeatability).
- **Structure (B49):** QD **core** → QD **shell** → **organic coating**.
- Bulk semiconductor: **continuous** levels within bands. QD: **discrete** levels and a **large** band gap (Schrödinger: confined electrons occupy quantised levels).
- **Release of confinement energy** (extra kinetic/band-edge energy of carriers squeezed into a tiny space) is a key QD property.
- Quantised gap → emission at a **very constant wavelength**, fine-tuned by changing dot size; energy released is consistent → **single-colour** emission.
- Larger dots (5–6 nm) → lower energy → red/orange. Smaller dots (2–3 nm) → higher energy → blue/violet.
- Band gap decreases as QD size increases; emission red-shifts with increasing size. Spectra (B46) span ~473 nm (blue) to 655 nm (red).
- Colour is "entirely dependent on size" — called **quantum confinement**.

QD synthesis B47–48	Key points
1 High-temperature (hot injection)	N_2 atmosphere, high-boiling solvent, heating mantle **> 200 °C**. Stage I substrate heating (low T) → Stage II precursor injection (high T) → Stage III reaction (high T). Route I: [A] precursor/solvent/ligand in flask, inject [B]. Route II: solvent/ligand in flask, inject [A]+[B] → AB nanocrystals.
2 Hydrothermal	Water as solvent at high T & pressure in a sealed steel autoclave (Teflon liner). Example: Bi_2Te_3 (dark grey powder) from $BiCl_3$ + Te + PVP + H_2O with NaOH, $NaBH_4$.
3 Microwave	Microwave-assisted (hydrothermal) synthesis of **fluorescent carbon dots from starch** (+ H_3PO_4); filtration → blue emission; liquid–liquid extraction → orange/green.
4 Ultrasonication	**Acoustic cavitation** in PEG-400 → metal-doped C-dots M@CDs (M = Ga, Sn, Zn, Ag, Au; 2.5 h, 65 °C); C-dots mixed with activated carbon → AC/C-dots.

QD applications B49–51	Uses
Bio/medical	Bioimaging (**cancer-cell imaging**), biosensing, immunoassays, drug delivery, detection of genetic disease, clinical use
Displays	Brighter, more vibrant colours — **QD-OLED TVs**
Solar cells	Absorb a **wider range of wavelengths** → higher efficiency
Lighting / devices	Solid-state lighting, **LEDs**, lasers, photocatalysts, batteries
Sensing & research	Metal-ion sensing (Ni^{2+}, Cd^{2+}, Zn^{2+}, Hg^{2+}, Cu^{2+}, Pb^{2+}, Ag^{+}, Mn^{2+}, Co^{2+}, Cr(VI)); sensors, quantum computing

Two different colour mechanisms: metal nanoparticles (Au, Ag) → **surface plasmon resonance**; quantum dots (semiconductors) → **quantum confinement / band gap**. Don't apply the QD rule (small = blue) to gold: 25 nm gold reflects **red**, 50 nm **green**, 100 nm **orange**.

4 · Formula sheet

Formula	Symbols & units	Conditions / notes	Slide
$V = IR$; $I = V/R$; $R = V/I$	V potential difference (V); I current (A); R resistance (Ω)	Constant T; ohmic material; shape/length unchanged	A3–4
$R = \rho L/A$	ρ resistivity (Ω·m); L length (m); A cross-sectional area (m^2)	$R \propto L$, $R \propto 1/A$	A4
$\rho = RA/L = VA/(IL)$	as above	Material + temperature only	A5
$\rho_T = \rho_0(1 + \alpha\Delta T)$	ρ_T at T; ρ_0 at reference T_0; α temperature coefficient (/°C); $\Delta T = T - T_0$	α > 0 metals; α < 0 semiconductors & insulators	A8
$\rho(T) = \rho_0$ $(T > T_c)$; 0 $(T < T_c)$	T_c critical temperature (K)	Superconductor; abrupt drop	A35
$H_c(T) = H_c(0)[1 - (T/T_c)^2]$	$H_c(T)$ critical field at T; $H_c(0)$ at 0 K; T operating temperature; T_c critical temperature	Type I. *$T = T_c \rightarrow H_c = 0$; $T = T_c/2 \rightarrow 0.75\ H_c(0)$*	A39
$\mu_L = -(e/2m_e)\ L$, $L = \sqrt{(l(l+1))}\ \hbar$	e electron charge; m_e electron mass; L orbital angular momentum; l orbital quantum number	Orbital magnetic moment	A48
$\mu_S = -g_s(e/2m_e)\ S$, $S = \sqrt{(s(s+1))}\ \hbar$	S spin angular momentum; s spin quantum number; **$g_s = 2$** (Landé g-factor)	Spin magnetic moment	A48
$\mu_B = e\hbar/2m_e = eh/4\pi m_e$	= **9.27×10^{-24} A·m²** or J/T	Unit of atomic moment	A48
$\mu_{eff} = \sqrt{(n(n+2))}\ \mu_B$	n = number of unpaired electrons	Spin-only (orbital quenched)	A48
$\chi = M/H$	M magnetization = moment per unit volume; H field strength (A/m)	χ < 0 dia; small > 0 para; ≫ 1 ferro	A49
$\mu = B/H$; $\mu = \mu_r\mu_0$; $\mu_r = \mu/\mu_0$	B flux density (T); H (A/m); μ (H/m); **$\mu_0 = 4\pi \times 10^{-7}$ H/m**	—	A49–50
$\mu_r = 1 + \chi$; $\mu = \mu_0(1 + \chi)$		Linear (B = μH) & isotropic only	A50
$\chi = C/T$ (Curie law)	C Curie constant; T absolute temperature	Paramagnets; 1/χ vs T line, slope 1/C	A55
$\chi = C/(T - T_C)$ (Curie–Weiss)	T_C Curie temperature	Ferromagnets **above** T_C	A58
$\lambda = h/mv$ (de Broglie)	h Planck's constant; m mass; v velocity	Confinement when particle size ~ λ	B18
Surface area of 1 cm^3 cut into n^3 cubes	6 → 12 → 18 cm^2 for n = 1, 2, 3	*general: $6n\ cm^2$*	B17
Chemical equations: sol-gel hydrolysis / water condensation / alcohol condensation (§3.5 table) · $YBa_2Cu_3O_7$, $Bi_2Sr_2Ca_2Cu_3O_{10}$, Nb_3Sn · Fe_3O_4, $BaFe_{12}O_{19}$, $BaO{\cdot}6Fe_2O_3$ · Al_2O_3 · SF_6			

5.1 · One-page cheat sheet

Resistance & resistivity

- V = IR (constant T, ohmic: Cu, Al). Non-ohmic: diodes, transistors.
- R = ρL/A. ρ depends on **material + T only**.
- ρ (Ω·m): conductors 10^{-8}–10^{-6} · semis 10^{-5}–10^{6} · insulators 10^{8}–10^{16}.
- Ag 1.6, Cu 1.7, Al 2.7 ($\times10^{-8}$); Fe 1.0×10^{-7}; Si ~10^{3}; glass 10^{10}–10^{14}; rubber ~10^{13}.
- α (/°C): Ag .0038, Cu .0037, Au .0034, **Al .00429**.
- 6 factors: material, temperature, impurities, strain, phase change, defects. In metals impurities, strain, defects ↑ ρ; annealing ↓ ρ.
- $\rho_T = \rho_0(1 + \alpha\Delta T)$: α + metals; − semis & insulators.
- Low α: Nichrome, Constantan → resistors, heaters. Negative α → thermistors.

Bands

- E_g: conductor ≈ 0 (overlap) · semi 0.7–1.5 (~1) eV · insulator > 5 eV.
- Diamond 5.5, glass 9, PVC/rubber ≫ 6 eV.
- Valence e^-: < 4 / = 4 / > 4. TCR: + / − / −.
- 0 K: VB full, CB empty for semis & insulators.

Cu vs Al

- Cu [Ar]$3d^{10}4s^{1}$; 2nd to Ag; 8.9 g/cm^3.
- Al [Ne]$3s^{2}3p^{1}$; 2.7 g/cm^3; 61 % of Cu; ⅓ weight (same size); ½ weight (same conductance); Al_2O_3; ACSR.
- Hard-drawn: strong, less conductive → overhead lines. Annealed: ductile, more conductive → house wiring.

Semiconductors

- Carriers: e^- (CB) + holes (VB). Doping ~1 in 10^{8}.
- n: P, As, Sb, (Bi) · donor → + ion · majority e^-.
- p: B, Al, Ga, In · acceptor → − ion · majority holes.
- GaAs: LEDs, laser diodes, microwave ICs. InP: high-speed, optical comm., 5G/6G.
- SCR, IGBT, Power MOSFET → converters, inverters, motor drives.

Superconductors

- 1911 Onnes, Hg 4.2 K · 1913 Nobel (liquefied He, ~1.5 K) · 1933 Meissner & Ochsenfeld · 1986 HTS Bednorz & Müller · 1987 Nobel + YBCO.
- LN_2 bp 77.355 K (−195.795 °C).
- Zero ρ below T_c + Meissner (χ = −1). Meissner separates SC from perfect conductor.
- Type I soft: complete Meissner, 1 low H_c, sharp — Pb, Hg, Al.
- Type II hard: vortices, H_{c1} (entry) & H_{c2} (destroyed), high current — Nb, NbTi, Nb_3Sn, YBCO, BSCCO.
- T_c (K): Hg 4.2, Pb 7.2, Nb 9.2, NbTi 10, Nb_3Sn 18, YBCO 92, BSCCO ~110.
- $H_c(T) = H_c(0)[1 - (T/T_c)^2]$.
- YBCO: ceramic cuprate, perovskite-like; CuO_2 planes; δ changes T_c; H_{c2} > 100 T; J_c ~10^6 A/cm^2; first T_c > 77 K.
- Josephson: 2 SC + thin barrier. SQUID: loop + 2 junctions, femtotesla, MEG. SMES stores energy in B-field. Maglev SCMaglev > 600 km/h. LHC, ITER, MRI, NMR. Cables: Essen.

Magnetism

- Moment = vector sum of orbital + spin. $\mu_B = e\hbar/2m_e = 9.27\times10^{-24}$ A m^2. $g_s = 2$.
- $\mu_{eff} = \sqrt{(n(n+2))}\,\mu_B$: 1.73, 2.83, 3.87, 4.90, 5.92 for n = 1–5.
- χ = M/H; μ = B/H; $\mu_0 = 4\pi\times10^{-7}$ H/m; $\mu_r = 1 + \chi$ (linear, isotropic).

	χ	μ_r	T-law
Dia	<0 small	<1	T-indep.
Para	>0 small	>1	C/T
Ferro	≫1	10^2–10^5	$C/(T-T_C)$ above T_C
Antiferro	>0 small	≈1	peak at T_N
Ferri	>0 mod–large	10^2–10^4	net ≠ 0

- Dia: Bi, Cu, Ag, Au, Hg, Zn, Si, P, water, graphite. Para: Al, Pt, Na, Sn, O_2, H_2, air. Ferro: Fe, Co, Ni, Gd, Dy, Er, Ho. Antiferro: MnO, FeO, CoO, NiO, MnS, MnF_2, FeMn. Ferri: Fe_3O_4, ferrites ($BaFe_{12}O_{19}$...).
- T_C (K): Co 1404, Fe 1043, Ni 631, Gd 293. T_N: Cr 308, Mn 100.
- Soft (high μ, low loss) → cores, mu-metal. Hard → NdFeB, SmCo. MO storage: heat above T_C. MRAM non-volatile.

Nano & QDs

- ≥ 1 dimension < 100 nm. 1 nm = 10^{-9} m; atom 0.1 nm; hair 50,000–100,000 nm.
- 0D QDs/NPs · 1D wires/tubes · 2D films · 3D bundles.
- Reasons: surface area ↑ (6→12→18 cm^2), quantum effects. Size ↓ → gap ↑.
- Au 25/50/100 nm → red/green/orange; nano-Au E_g 3.4 eV, mp 940 vs 1080 °C. SPR.
- Top-down: milling, etching, laser ablation, sputtering, electro-explosion.
- Milling: 50 µm, WC/steel balls, 5:10, 100–200 °C, Fe < 1–2 %, < 300 ppm, > 30 h → > 10 %.
- Sol-gel: alkoxide; hydrolysis → condensation → aging → drying → calcination. Aero = supercritical; xero = evaporation.
- QD 2–10 nm, 100–1000 e^-, 10–50 atoms; CdSe, CdS, InP; 5–6 nm red, 2–3 nm blue.

5.2 · Common MCQ traps & easily confused pairs

1. **Resistance vs resistivity:** only R depends on L and A; ρ depends on material + temperature.
2. Ohm's law fails for **diodes and transistors** and when temperature changes.
3. Heating: metal ρ **rises**; semiconductor **and insulator** ρ **falls**.
4. Impurity in a metal always **raises** ρ (even a lower-ρ impurity); doping a semiconductor **lowers** ρ.
5. Annealing lowers ρ → annealed Cu more conductive but weaker; hard-drawn for overhead lines.
6. Low α → Nichrome/Constantan (resistors, heaters); negative α → thermistors; high α metals unsuitable for precise resistors.
7. Lowest ρ: **Ag** (Cu second). Highest α among Ag, Cu, Au, Al: **Al**.
8. Al: most abundant **metal**, 3rd most abundant **element**. 61 % of Cu's conductivity.
9. Valence electrons: conductor < 4, semiconductor = 4, insulator > 4.
10. Holes live in the **valence** band; free electrons in the **conduction** band.
11. Pentavalent → n-type → donor → **positive** ion. Trivalent → p-type → acceptor → **negative** ion.
12. Al is a conductor, a **p-type dopant**, a **Type I** superconductor and a **paramagnet** — all in these slides.
13. GaAs → LEDs/laser diodes/microwave ICs; InP → optical communication/high speed. InP is also a QD material.
14. Meissner effect, not zero resistance, distinguishes a superconductor from a perfect conductor.
15. Superconductor $\chi = -1$ (perfect); ordinary diamagnet χ only slightly negative.
16. H_{c1} = flux **starts entering**; H_{c2} = superconductivity **destroyed** (Type II only).
17. Nb (element) is **Type II**; Pb, Hg, Al are Type I.
18. Dates: discovery 1911; Onnes Nobel 1913; Meissner 1933; HTS 1986; HTS Nobel 1987 (= year of YBCO).
19. YBCO 92 K beats liquid N_2 (77 K) → no liquid He needed. BSCCO (~110 K) is the highest T_c in the table.
20. In YBCO, superconductivity lives in the **CuO_2 planes**; oxygen deficiency δ tunes T_c.
21. SQUID = **two** Josephson junctions in a superconducting loop.
22. T_c = critical temperature (superconductor) vs T_C = Curie temperature (ferromagnet).
23. Curie law C/T → paramagnets; Curie–Weiss C/(T − T_C) → ferromagnets above T_C.
24. Para χ falls with T; antiferro χ **rises** to T_N then falls; dia χ is T-independent; at T_C ferro $\chi \to \infty$.
25. Antiferro: equal antiparallel → net 0, $\mu_r \approx 1$. Ferri: unequal antiparallel → net ≠ 0.
26. Fe ferro · FeO antiferro · Fe_3O_4 ferri.
27. Cu, Ag, Au, Bi, water, graphite = **dia**; Al, Pt, Na, O_2 = **para**.
28. Elements Cr and Mn are antiferromagnetic (T_N only). Tb–Tm have both T_C and T_N, with $T_N > T_C$.
29. Highest Curie temperature: **Co (1404 K)**, not Fe (1043 K). Gd 293 K ≈ room temperature.
30. μ_{eff} counts **unpaired** electrons only; net moment is a **vector** sum.
31. Magneto-optical rewriting heats above the **Curie** temperature. MRAM is **non-volatile**.
32. Soft magnets → transformer/motor cores, shielding (mu-metal). Hard magnets → NdFeB, SmCo permanent magnets.
33. Nanomaterial needs only **one** dimension < 100 nm.
34. Dimensionality counts dimensions **outside** nanoscale: QD 0D, nanotube 1D, thin film 2D.
35. Cutting a cube keeps volume constant but raises area → better **catalysis**.
36. Gold nanoparticles: 25 nm **red**, 50 nm **green**, 100 nm **orange**. Silver 100 nm sphere yellow vs 100 nm **prism** red (shape matters).
37. Nano gold is a semiconductor (3.4 eV), a good catalyst, and melts **lower** (~940 °C) than bulk (~1080 °C).
38. Metal-NP colour = SPR; QD colour = quantum confinement.
39. Laser **ablation**, sputtering, etching, electro-explosion = top-down. Laser **pyrolysis**, spinning, CVD, sol-gel, chemical reduction, green synthesis, supercritical fluid, molecular condensation = bottom-up.
40. Milling a.k.a. mechanical alloying / mechanochemical synthesis; adv = high production rate; big problem = contamination.
41. Sol-gel: adv = low temperature, monosized particles; disadv = slow, agglomeration, leftover reactants/growth aids.
42. Sol-gel precursor = **metal alkoxide**; aging lowers porosity but increases particle spacing.
43. Water condensation releases **H_2O**; alcohol condensation releases **ROH**; reverse of hydrolysis = re-esterification; reverse of alcohol condensation = alcoholysis.
44. Aerogel = supercritical drying; xerogel = ordinary evaporation.
45. QD: bigger → smaller gap → red (lower energy); smaller → bigger gap → blue. Emission red-shifts as size grows.
46. Confinement occurs when size ~ **de Broglie wavelength** ($\lambda = h/mv$), not the wavelength of light.
47. Hydrothermal = **water** + high T & P in a sealed **autoclave**; hot injection > 200 °C under N_2.

5.3 · Where the slides disagree — what to answer

Topic	What the slides say	Safest answer
Insulator resistivity	A5 & A28: 10^8–10^{16} Ω·m · A7: 10^{10}–10^{14} · A16 table: 10^{12}–10^{18}	10^8–10^{16} (stated twice); recognise the others if offered alone
Semiconductor resistivity	A5: 10^{-5}–10^6 · A16 table: 10^0–10^2	10^{-5}–10^6; pick whichever range appears
Semiconductor E_g	A13: ~1 eV · A15/A30: 0.7–1.5 eV · A17 table: < 5 eV	0.7–1.5 eV (~1 eV)
Semiconductor at room T	A15: "behaves as an insulator at room temperature or low temperature" · A30: gains conductivity at room temperature	Insulator at **0 K**; small conductivity at room T
Magnesium	A51 table: paramagnetic · A52: diamagnetic	Avoid relying on Mg; if forced, note both appear
QD size vs frequency	B44: "positive relationship between QD size and frequency" · B45–46: smaller → higher energy (blue); gap falls as size rises	**Inverse**: larger dot → lower frequency (red). B44 wording is misleading
Sol-gel condensation equations	B35 pairs the H_2O-releasing Si equation with "MOH + ROM → MOM + ROH" and vice versa · B34 pairs them correctly	OH + HO → H_2O (water condensation); OR + HO → ROH (alcohol condensation)
First T_c > 77 K	A34: HTS of 1986 had T_c above 77 K · A40: YBCO (1987) was the first with T_c > 77 K	"First with T_c > 77 K / LN_2-cooled" → **YBCO**; "HTS discovered" → 1986, Bednorz & Müller
Bohr magneton	A48 calls $e/2m_e$ "the Bohr magneton", then defines $\mu_B = e\hbar/2m_e$	$\mu_B = e\hbar/2m_e = 9.27 \times 10^{-24}$ A·m²
Nanoparticle light	B14: "scatter visible light rather than absorb" · same slide: SPR absorbs specific wavelengths	SPR: some wavelengths absorbed, rest scattered/reflected

Antiferro ordering temp.	A64 summary writes "below T_c" for antiferromagnets · A61: Néel temperature	**T_N** (Néel)
Donor examples	A31: P, As, Sb · A32: As, Sb, Bi, P	All four are donors
Copper bands	A21 labels 4s as VB and 3d as CB	Key point: bands overlap → $E_g \approx 0$
Bulk gold m.p.	B15: ~1080 °C	Use the slide value ~1080 °C (vs ~940 °C nano)

5.4 · Coverage checklist

☐	Topic	Subtopics	Slides	Here
☐	Ohm's law, resistance, resistivity	V = IR, validity, non-ohmic devices, R = ρL/A, ρ ranges & values	A3–A5, A7	1.1
☐	Factors affecting resistivity	Material, temperature, impurities, strain, phase change, defects, annealing	A6–A10, A19	1.2
☐	Temperature coefficient	$\rho_T = \rho_0(1 + \alpha\Delta T)$, sign of α, Nichrome/Constantan, thermistors, A20 table	A8, A20	1.3
☐	Energy band theory	VB, CB, forbidden gap, E_g values	A12–A13	1.4
☐	Conductor / semiconductor / insulator	10-row comparison, valence electrons, TCR, examples	A15–A17	1.5
☐	Conductors: Cu, Al	Required properties, Cu config & uses, hard-drawn vs annealed, Al facts, ACSR, Cu vs Al	A19–A27	1.6
☐	Insulators	E_g values, properties, materials, applications	A28	1.7
☐	Semiconductors	Carriers, intrinsic, extrinsic n/p, compound (GaAs, InP), n vs p table, applications	A30–A33	1.8
☐	Superconductors	History, zero resistivity, Meissner, Type I/II, T_c table, $H_c(T)$, J_c, YBCO, applications	A34–A43	1.9
☐	Magnetic moment	Orbital/spin moments, μ_B, g_s, μ_{eff}	A45–A48	2.1
☐	χ and μ	Definitions, units, μ_0, $\mu_r = 1 + \chi$, linear/isotropic	A49–A50	2.2
☐	Dia, para, ferro, antiferro, ferri	χ, μ_r, examples, Curie law, Curie–Weiss, T_C, T_N, domains, hysteresis, spin diagrams, A62 table	A47, A51–A64	2.3
☐	Applications & storage devices	Soft/hard magnets, MRI, HDD, magneto-optical, MRAM	A65–A66	2.3
☐	Liquid cooling	Coolant types, thermal conductivity, applications — **no slides**	syllabus only	6
☐	Nanomaterials intro	Definition, scale, uses, classification 0D–3D, categories	B2–B10	3.1–3.2
☐	Why nano is different	Surface area/volume, quantum size effect, confinement, de Broglie, other changes	B11, B17–B19	3.3
☐	Bulk vs nano gold	Property table, Au/Ag colours, SPR	B12–B15	3.4
☐	Top-down & bottom-up	Definitions, method lists, mechanical milling, sol-gel, reactions, xerogel/aerogel, ZnO example	B20–B40	3.5
☐	Quantum dots	Definition, mechanism, properties, colour vs size, spectra, synthesis (4 methods), structure, applications	B41–B51	3.6

6 · Liquid cooling — in your syllabus, but on no slide

Gap: the Module 3 syllabus box (A1, B1) lists "*Liquid cooling – types of coolants, thermal conductivity, and applications*", but **neither deck has a single slide on it**. Questions are unlikely if the quiz is set strictly from these slides. The points below are **standard textbook background, not from your slides** — use them only as a safety net.

- **Liquid cooling:** a circulating liquid absorbs heat from a hot component and rejects it at a radiator / heat exchanger. Liquids beat air because of higher thermal conductivity and heat capacity.
- **Thermal conductivity (k):** ability to conduct heat; Fourier's law $q = -k(dT/dx)$; unit **$W\ m^{-1}\ K^{-1}$**. Water ≈ 0.6 vs air ≈ 0.026 $W\ m^{-1}\ K^{-1}$.
- **Coolant types:** water (deionised) · water–glycol (antifreeze; lowers freezing point) · dielectric fluids (mineral/synthetic oils, fluorinated fluids) for direct/immersion cooling of electronics · liquid metals · refrigerants (two-phase) · cryogens (liquid N_2 77 K, liquid He — link to superconductors, A34/A40).
- **Good coolant:** high specific heat (water ≈ 4.18 $J\ g^{-1}\ K^{-1}$) · high k · low viscosity · low freezing / high boiling point · chemically stable, non-corrosive, non-toxic, non-flammable · electrically insulating if it touches electronics · cheap.
- **Applications:** CPU/GPU cold plates, data-centre immersion cooling, EV battery & motor cooling, engine radiators, transformer oil, nuclear reactors, MRI/superconducting magnets (liquid He / N_2).